

A geometric and Bayesian formulation of autoregressive language models

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Abstract

The mechanics of autoregressive large language models (LLMs) are formalized by re-framing the forward pass and pre-training objectives through the lens of Bayesian inference, high-dimensional geometry, and statistical mechanics. The conditional probability distribution generated by the Transformer architecture is shown to be explicitly decomposable into an intrinsic token prior and a contextual likelihood. These probabilistic components are mapped directly onto the structural blocks of the network—specifically multi-head attention (MHA) and feed-forward networks (FFN). It is demonstrated that the final output token distribution is modulated by the statistical variance of the latent residual stream, which acts as a dynamic inverse temperature. Finally, pre-training is formalized as a self-organizing geometric optimization by which these components are resolved into a stable semantic coordinate system.

1 Introduction and the Bayesian framing

An autoregressive large language model operates over a finite vocabulary V [2]. Given a context sequence of n tokens, $C = (t_1, t_2, \dots, t_n)$, where $t_i \in V$, a discrete conditional probability distribution over the vocabulary is defined for the next token $w \in V$:

$$P(w \mid C) \tag{1}$$

While this distribution is directly evaluated by modern deep learning architectures using discriminative causal language modeling [3], the formulation can be inverted using Bayes' theorem to expose the underlying generative dynamics [9]:

$$P(w \mid C) = \frac{P(C \mid w) \cdot P(w)}{P(C)} \tag{2}$$

Since the denominator $P(C) = \sum_{v \in V} P(C \mid v)P(v)$ represents the marginal likelihood of the context sequence, it serves as a constant normalization factor across all candidate tokens at step $n + 1$. The relative probability mass assigned to any token w is therefore determined strictly by the numerator:

$$P(w \mid C) \propto P(C \mid w) \cdot P(w) \tag{3}$$

where:

1. $P(w)$ denotes the **unconditional prior probability** of token w occurring in the language, independent of local context.
2. $P(C \mid w)$ represents the **contextual likelihood**, measuring the alignment of the historical sequence C with the trailing token w .

2 Mathematical decomposition of likelihood $P(C \mid w)$

In a Transformer architecture [1], the context sequence C is transformed into a sequence of continuous vectors. The input embedding matrix for the context is denoted as $X \in \mathbb{R}^{n \times d}$, where d is the model dimension. The final hidden state at the terminal position n after L layers is denoted as $h_n^{(L)} \in \mathbb{R}^d$.

The unnormalized log-probabilities (logits) z_w for each token w are computed by projecting $h_n^{(L)}$ onto an unembedding matrix $U \in \mathbb{R}^{|V| \times d}$, such that $z_w = h_n^{(L)} \cdot u_w$, where $u_w \in \mathbb{R}^d$ is the unembedding vector of the target token [4].

The log-likelihood $\log P(C \mid w)$ is shaped by the interaction of two foundational architectural components within each layer l : multi-head attention (MHA) and the position-wise feed-forward network (FFN).

2.1 Multi-head attention as contextual routing

The multi-head attention block at layer l and position n computes a dynamically weighted combination of all preceding token states [1]. The number of attention heads is denoted by H , and the head dimension by d_k . For a single attention head $g \in \{1, \dots, H\}$, the query, key, and value projection matrices are denoted by $W_Q^{(l,g)}, W_K^{(l,g)}, W_V^{(l,g)} \in \mathbb{R}^{d \times d_k}$, respectively.

The terminal query vector for head g is defined as $q_n^{(l,g)} = h_n^{(l-1)} W_Q^{(l,g)}$, whereas the keys and values for all historical positions $i \in \{1, \dots, n\}$ are represented as $k_i^{(l,g)} = h_i^{(l-1)} W_K^{(l,g)}$ and $v_i^{(l,g)} = h_i^{(l-1)} W_V^{(l,g)}$. The dynamic attention weights $\alpha_{n,i}^{(l,g)}$, which formalize conditional token-to-token dependencies, are expressed as:

$$\alpha_{n,i}^{(l,g)} = \frac{\exp\left(\frac{q_n^{(l,g)} \cdot k_i^{(l,g)}}{\sqrt{d_k}}\right)}{\sum_{j=1}^n \exp\left(\frac{q_n^{(l,g)} \cdot k_j^{(l,g)}}{\sqrt{d_k}}\right)} \quad (4)$$

The concatenated outputs of all heads are projected back into the residual stream via the global output matrix $W_O^{(l)} \in \mathbb{R}^{d \times d}$. By partitioning $W_O^{(l)} = [(W_{O,1}^{(l)})^T \dots (W_{O,H}^{(l)})^T]^T$, where each head's output projection slice is represented as $W_{O,g}^{(l)} \in \mathbb{R}^{d_k \times d}$, the total attention contribution to the log-likelihood of a target token w is decomposed into a sum of bilinear forms:

$$\log P(C \mid w)_{\text{MHA}} \sim \sum_{g=1}^H \sum_{i=1}^n \alpha_{n,i}^{(l,g)} \left(h_i^{(l-1)} W_V^{(l,g)} W_{O,g}^{(l)} u_w \right) \quad (5)$$

Two distinct mechanisms are thereby isolated: contextual routing is determined by $\alpha_{n,i}^{(l,g)}$, whereas semantic alignment between the historical features of token i and target token w is mediated by the bilinear inner product.

2.2 Feed-forward networks as concept memory retrieval

Following the attention block, the updated hidden state $x_n^{(l)} = h_n^{(l-1)} + a_n^{(l)}$ is processed by the position-wise feed-forward network (FFN). The FFN is modeled as a key-value associative memory [5]. The parameters are represented by $W_1^{(l)} \in \mathbb{R}^{d \times d_{ff}}$ (denoting intermediate keys $\{k_m^{(l)}\}_{m=1}^{d_{ff}}$) and $W_2^{(l)} \in \mathbb{R}^{d_{ff} \times d}$ (denoting intermediate values $\{v_m^{(l)}\}_{m=1}^{d_{ff}}$).

Upon application of a non-linear activation function σ (e.g., GELU [6]), the operation at the terminal position is formulated as:

$$\text{FFN}(x_n^{(l)}) = \sum_{m=1}^{d_{ff}} \sigma \left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right) v_m^{(l)} \quad (6)$$

Projection of this operation onto the target token unembedding vector u_w yields the FFN likelihood formulation:

$$\log P(C \mid w)_{\text{FFN}} \sim \sum_{m=1}^{d_{ff}} \sigma \left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right) \cdot (v_m^{(l)} \cdot u_w) \quad (7)$$

The activation term $\sigma \left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right)$ scales the degree of satisfaction of the m -th latent conceptual rule by the synthesized context, whereas the term $(v_m^{(l)} \cdot u_w)$ represents the semantic correlation between that rule and the vocabulary token w .

3 Geometric formulation of the prior $P(w)$ and logit mechanics

The exact computation of the unconditional prior $P(w)$ requires marginalization over the space of all possible context sequences $\mathcal{C}$:

$$P(w) = \sum_{C \in \mathcal{C}} P(w \mid C) P(C) \quad (8)$$

This expectation is statically approximated within the parameters of the Transformer via two geometric mechanisms:

3.1 Unembedding bias (b_U)

In architectures featuring an explicit unembedding bias $b_U \in \mathbb{R}^{|V|}$, the logit calculation is defined as $z_w = (h_n^{(L)} \cdot u_w) + b_{U,w}$. Under an uninformative context where $h_n \rightarrow \mathbf{0}$, the logit collapses directly to the scalar bias component, yielding:

$$P(w) \approx \frac{\exp(b_{U,w})}{\sum_{v \in V} \exp(b_{U,v})} \quad (9)$$

3.2 Embedding norms ($\|u_w\|$)

In bias-free architectures, the prior is encoded in the Euclidean magnitude of the token vector, $\|u_w\|$. Geometric decomposition of the final inner product yields:

$$z_w = \|h_n^{(L)}\| \|u_w\| \cos(\theta_w) \quad (10)$$

where θ_w is the angle between the final context state and the token vector. Longer vector magnitudes $\|u_w\|$ are developed during optimization by tokens with high baseline frequencies in the training corpus, thereby amplifying the baseline logit irrespective of the contextual direction $\cos(\theta_w)$ [7].

4 Statistical mechanics of the latent stream

The remaining term in the geometric logit decomposition is the norm of the final hidden state, $\|h_n^{(L)}\|$. This component serves as a dynamic, context-dependent entropy controller.

4.1 Activation variance as inverse temperature

The final hidden state $h_n^{(L)}$ is represented as a row vector $h_n = [x_1, x_2, \dots, x_d] \in \mathbb{R}^d$. If the individual channel activations are treated as samples from an internal distribution with a mean driven to zero by layer normalization ($\mu_x \approx 0$) [8], the cross-channel variance is defined as:

$$\sigma^2 = \frac{1}{d} \sum_{i=1}^d (x_i - \mu_x)^2 \approx \frac{1}{d} \sum_{i=1}^d x_i^2 \quad (11)$$

Since the squared Euclidean norm is given by $\|h_n\|^2 = \sum_{i=1}^d x_i^2$, the vector magnitude is expressed as a function of the activation variance:

$$\|h_n\| = \sqrt{d \cdot \sigma^2} \quad (12)$$

Upon substitution of this relation into the vocabulary-wide Softmax function, $\|h_n\|$ is mapped to the inverse temperature parameter $\beta = \frac{1}{T}$ from statistical mechanics [9, 10]:

$$P(w \mid C) = \frac{\exp(\beta \cdot \|u_w\| \cos(\theta_w))}{\sum_{v \in V} \exp(\beta \cdot \|u_v\| \cos(\theta_v))}, \quad \text{where } \beta = \|h_n\| \quad (13)$$

1. **High contextual certainty:** In predictable contexts, aligned information is written into the residual stream by successive attention heads and FFN layers. The cross-channel variance σ^2 is inflated via constructive interference, causing $\|h_n\|$ to expand. A high inverse temperature β is yielded, thereby sharpening the distribution and lowering entropy.
2. **High contextual ambiguity:** In entropic or transitional contexts, orthogonal or destructive vectors are written into the residual stream by architectural sub-layers. The cross-channel variance σ^2 is collapsed, shrinking $\|h_n\|$. A low inverse temperature β is yielded, flattening the distribution into a high-entropy state.

4.2 Cosine similarity as a correlation coefficient

Similarly, by treating the dimensions of h_n and u_w as zero-mean random variables, the likelihood component $\cos(\theta_w)$ is mapped directly to the Pearson correlation coefficient (r_{h_n, u_w}) in feature space:

$$\cos(\theta_w) = \frac{\text{Cov}(h_n, u_w)}{\sigma_{h_n} \sigma_{u_w}} = r_{h_n, u_w} \quad (14)$$

where σ_{h_n} and σ_{u_w} denote the standard deviations of the components of h_n and u_w , respectively. The unnormalized logit is therefore proportional to the total feature covariance scaled by the model dimension: $z_w = \text{Cov}(h_n, u_w) \cdot d$.

5 Statistical description of pre-training

While unembedding vectors u_w remain static during inference, both the context state h_n and the vocabulary matrix U are optimized simultaneously during pre-training via minimization of the cross-entropy loss [11] for a target next-token w^* :

$$\mathcal{L} = -\log \frac{\exp(h_n \cdot u_{w^*})}{\sum_{v \in V} \exp(h_n \cdot u_v)} \quad (15)$$

This optimization framework is modeled as a dual-force geometric process:

5.1 The mutual attraction force

The gradient with respect to the correct token vector u_{w^*} is interpreted as an attractive force:

$$\frac{\partial \mathcal{L}}{\partial u_{w^*}} = -(1 - P(w^* | C))h_n \quad (16)$$

By this force, u_{w^*} and the context representation h_n are dynamically rotated toward one another, thereby maximizing the directional correlation $\cos(\theta_{w^*})$. Simultaneously, for a token w^* consistently validated across diverse contexts, the vector norm $\|u_{w^*}\|$ is driven upward to minimize corpus-wide loss, thereby establishing the global prior $P(w)$.

5.2 The global repulsion force

Conversely, a contrastive repulsion force against all incorrect tokens $v \neq w^*$ is generated by the denominator of the Softmax, resembling contrastive representation learning frameworks [12]:

$$\frac{\partial \mathcal{L}}{\partial u_v} = P(v | C)h_n \quad (17)$$

The context state h_n is thereby pushed away from incorrect vocabulary vectors, forcing them toward orthogonality ($\cos(\theta_v) \rightarrow 0$).

5.3 Emergence of distributional semantics

Through loss minimization over a large corpus, semantic clustering emerges. For distinct tokens w_1 and w_2 sharing high contextual substitutability, identical directional updates are repeatedly received from shared context states h_n . A geometric steady-state is thereby established by this self-organizing system, wherein a coordinate system is discovered by vocabulary vectors such that spatial proximity mirrors latent semantic relationships, validating the distributional hypothesis [13]:

$$\cos(\theta_{w_1, w_2}) \rightarrow 1 \iff \mathcal{C}_{w_1} \approx \mathcal{C}_{w_2} \quad (18)$$

where $\mathcal{C}_w$ represents the distribution of contexts in which token w appears.

6 Conclusion

By formalizing the large language model as a Bayesian engine, a transition is made from a qualitative understanding of neural network behaviors to an exact geometric framework. A context vector is iteratively constructed in tandem by multi-head attention and feed-forward networks, where the final magnitude represents the statistical certainty (β) of the model, and the orientation represents structural feature configurations. The final token prediction is therefore represented as a direct measure of the Pearson correlation between this dynamic contextual variance and the self-organized coordinate landscape of the vocabulary embeddings.

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